

NVT Data Analysis Task 7

Business Intelligence Dashboard Using Power BI

Prepared using the supplied train.csv dataset.

1. Project Objective

The objective is to transform a sales dataset into an interactive Business Intelligence dashboard using Power BI, Power Query and basic DAX. The dashboard is designed to present key sales KPIs, trends, category performance and regional performance.

2. Dataset Overview

Item	Value
Rows	9,800
Original columns	23
Unique orders	4,922
Unique customers	793
Date range	2015-01-02 to 2018-12-11
Missing Postal Codes	11

3. Data Preparation

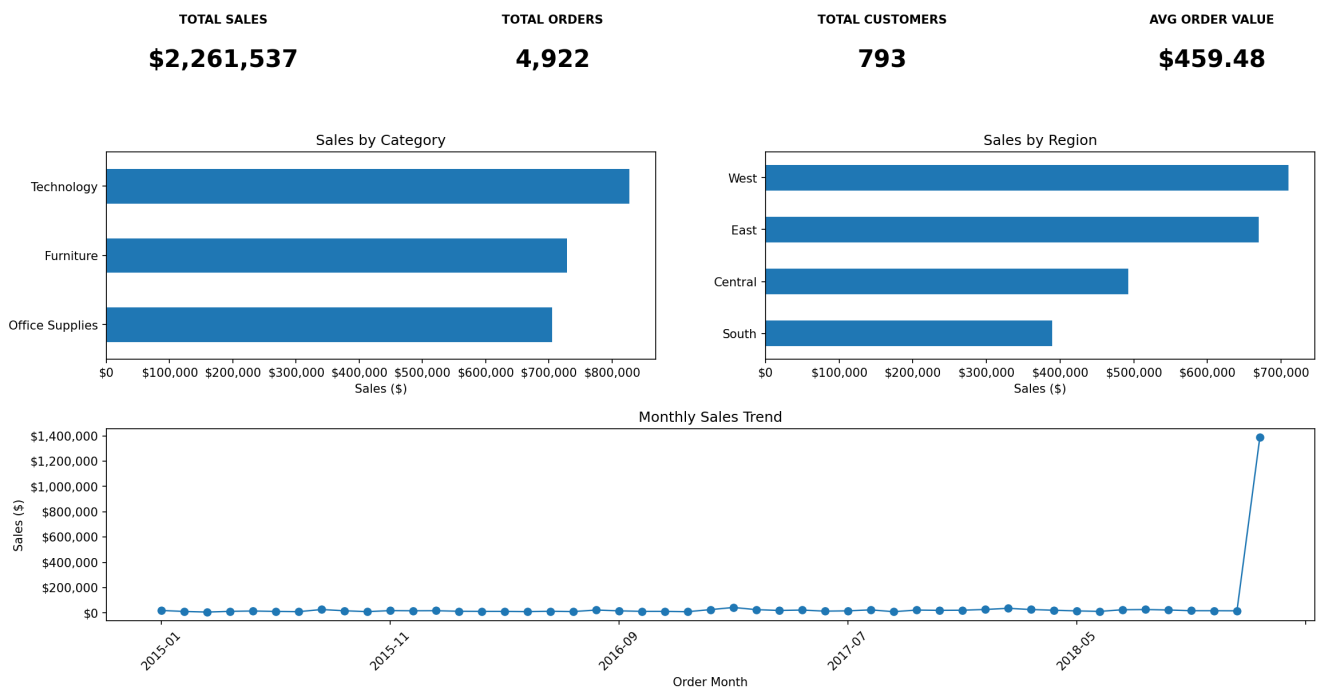
Order Date and Ship Date were parsed as US-style month/day/year dates. Postal Code was retained as a text-compatible field and missing postal codes were labeled Unknown in the Power BI-ready file. Additional helper fields were created for Order Year, Order Month, Order Month Name, Order Year-Month and Shipping Days.

4. Dashboard KPIs

KPI	Actual value
Total Sales	\$2,261,536.78
Total Orders	4,922
Total Customers	793
Average Order Value	\$459.48
Average Shipping Days	107.01

5. Dashboard Visuals

SUPERSTORE SALES & BUSINESS INTELLIGENCE DASHBOARD



The image above is a prepared dashboard-style visual based on the uploaded dataset. The native interactive version should be recreated and saved as a .pbix file in Power BI Desktop.

6. Business Insights

- The highest-sales category is Technology.
- The highest-sales region is West.
- The highest-sales segment is Consumer.
- The strongest sales month in the dataset is NaT, with sales of \$1,389,173.66.
- The highest-sales sub-category is Phones.
- The average shipping duration is 107.01 days.

7. DAX Measures

Recommended measures are Total Sales, Total Orders, Total Customers, Average Order Value, Average Shipping Days and Sales YTD. The dataset does not contain Profit or Quantity, so Profit Margin and Quantity-based KPIs are intentionally excluded.

8. Conclusion

The completed solution demonstrates data preparation, KPI design, interactive visualization planning and business insight generation using Power BI. The supplied supporting files make the project ready for final construction in Power BI Desktop and GitHub submission.